

NOISY REWARDS WRITE NOISY MEMORIES: REWARD ERROR AS A VARIANCE AMPLIFIER IN SELF-IMPROVING AGENTS

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Paper under double-blind review

ABSTRACT

Self-improving agents increasingly turn terminal reward into persistent natural-language memory. This creates a state-update channel that is easy to overlook. An evaluator error can change the interpretation assigned to an otherwise identical trajectory. That altered interpretation is stored and later reused. We isolate this channel in the released ReasoningBank implementation using WebArena Shopping trajectories. Among the four paired interventions that complete in both writer conditions, flipping only the reward-conditioned reflection mode changes every written memory, with mean token-set Jaccard distance 0.735. A disjoint eight-trajectory control rules out a generic prompt-wording explanation: reward-mode memory distance is 0.700 versus 0.595 under stronger semantic-preserving same-mode rewordings, giving paired excess 0.105 and exact $p = 0.0078$. The perturbation persists into future behavior. A fixed-state action probe changes the next structured action in 7 of 12 aligned pairs. An independently frozen fixed-evidence terminal replication then crosses every complete source-memory pair with every frozen future task. Across 256 fresh rollouts, the mean absolute success-rate difference is 0.15625 and a 100,000-draw within-cell permutation test gives $p = 0.00074$. A plug-in decomposition of these conditional effects yields an additional between-memory success-variance term of $0.0762q(1 - q)$ under reward-label corruption probability q . In this released configuration, evaluation error can become durable agent state and measurably alter later outcome distributions before any parameter update occurs. The result identifies reward reliability as a state-consistency requirement for memory-based self-improvement systems in which terminal outcomes choose what experience is stored.

1 INTRODUCTION

Self-improving agents often treat the end of one episode as the beginning of the next learning update. In external-memory systems, this update can happen without gradients. The agent summarizes a completed trajectory and stores the resulting memory. Later tasks retrieve that memory as context. The terminal evaluator therefore does more than score behavior. When its output controls memory construction, the evaluator also controls persistent agent state.

Evaluator unreliability is already well documented: JudgeBench and RewardBench expose disagreement and failure modes in LLM judges and reward models (Tan et al., 2025; Lambert et al., 2024). The released ReasoningBank implementation makes the downstream state channel explicit (Ouyang et al., 2026): score one selects a success-reflection writer, while score zero selects a failure-reflection writer over the same trajectory. Reflexion likewise turns evaluative feedback into persistent verbal state (Shinn et al., 2023). Thus our novelty is not that judges can be wrong or that outcome-conditioned text can be stored; it is the matched write-time intervention created when an erroneous terminal bit changes persistent state while the source experience is held fixed. Figure 1 shows this write-and-reuse channel.

This channel sits upstream of recent memory-valuation results. Memory Reward Inflation shows that proxy reward can inflate erroneous memories during reuse (Asadolahi et al., 2026); RoMeRL studies trajectory-level utility updates and the memory-reward trap (Yang et al., 2026); and the agent-fragility

study measures run variance, task-order sensitivity, and memory underspecification while deliberately using ground-truth reward in its main memory experiments (Ye et al., 2026). These works cover valuation, reuse, and downstream variance. Our residual object is earlier: whether reward error changes *what is written* from an identical experience, and whether this write-time state intervention propagates under matched future evidence.

A naive test is insufficient because the success and failure writers use different instructions by construction. Different prompts can produce different text even when the reward semantics are irrelevant. We therefore treat generic prompt sensitivity as the strongest simple reduction. Our control uses eight fresh trajectories and semantic-preserving same-mode rewordings whose lexical prompt perturbation is larger than the released success-versus-failure prompt difference. Reward-mode memory divergence remains larger: 0.700 versus 0.595, with paired excess 0.105 and exact $p = 0.0078$.

The underlying write intervention is equally direct. We hold each released trajectory byte-for-byte fixed and flip only the reward-conditioned reflection mode. Among the four requested pairs that complete in both writer conditions, every pair changes normalized memory content and memory-item titles. Mean token-set Jaccard distance is 0.735, and a deterministic strategy-slot audit over the frozen titles also changes in 4/4 complete pairs. Two additional failure-label calls terminate at the provider response-state layer before assistant text exists, so the 4/4 statement is explicitly conditional on paired completion. The complete-pair effect occurs for source trajectories from both original-outcome strata, so it is not an artifact of drawing only failed episodes.

We then carry the intervention across the episode boundary. A fixed-state action probe changes the next structured browser action in 7 of 12 aligned paired rollouts. The decisive downstream test freezes all four complete source-memory pairs and all four future tasks before additional policy calls. Every source memory is crossed with every future task. Eight fresh rollouts are executed per memory condition in each of the 16 cells. All 256 calls complete. Mean absolute terminal success-rate difference is 0.15625, and a 100,000-draw within-cell permutation test gives $p = 0.00074$, passing the frozen effect and significance gates.

The resulting picture is a causal decomposition of self-improvement variance. Reward changes the memory-writing objective while the source experience stays fixed. The written state then reaches future decisions under matched evidence. Repeated memory-conditioned rollouts change terminal outcome distributions. The measured conditional effects yield an explicit between-memory variance component under stochastic reward-label corruption. This establishes evaluator reliability as a state-consistency problem for memory-based self-improvement, not only an endpoint-scoring problem.

Contributions and closest-work boundary. We make three bounded contributions. First, we isolate reward-conditioned memory construction by holding the source trajectory byte-identical and flipping only writer mode. Second, we challenge generic prompt sensitivity with a stronger same-mode wording control. Third, we carry the paired memory state across the episode boundary into fixed-state action probes and a fully crossed 256-rollout terminal test. The paper does *not* claim the first reward-robustness benchmark, the first outcome-conditioned memory writer, or the first memory-reward amplification result; its claim is the causal write-time channel and its downstream consequences in this frozen ReasoningBank/WebArena configuration.

2 REWARD ERROR AS A PERSISTENT-STATE INTERVENTION

Let task x produce trajectory τ , and let the terminal evaluator emit $r \in \{0, 1\}$. A reward-conditioned memory writer constructs

$$M(r) = G(\tau, r). \quad (1)$$

In the released ReasoningBank implementation, $r = 1$ selects the success-reflection objective and $r = 0$ selects the failure-reflection objective. The intervention studied here holds τ fixed and changes only this branch. The immediate causal estimand is therefore the paired persistent-state contrast $M(1)$ versus $M(0)$ for the same experience.

For a matched future task state x' , let Y denote a downstream behavioral or terminal outcome under retrieved memory M . The reuse-stage contrast is

$$\Delta_Y(x') = \mathbb{E}[Y \mid do(M = M(0)), x'] - \mathbb{E}[Y \mid do(M = M(1)), x']. \quad (2)$$

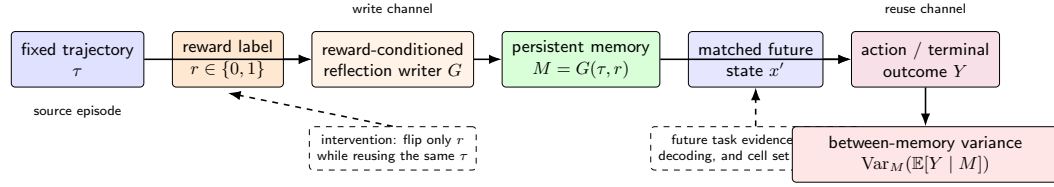


Figure 1: The reward-to-memory-to-outcome channel. The source trajectory is fixed while the reward label selects the reflection writer. The resulting persistent memory is then injected into a matched held-out task packet under a fixed policy configuration. Repeated outcomes identify the between-memory component of run variance.

The future task evidence and policy configuration are held fixed across the pair. This separates variation introduced by persistent memory state from variation in the observed future state.

2.1 VARIANCE AMPLIFICATION

Across self-improvement runs, reward errors can cause different memories to be written from otherwise comparable experiences. Let M denote the persistent memory state available to later tasks. The law of total variance gives

$$\text{Var}(Y) = \mathbb{E}_M[\text{Var}(Y | M)] + \text{Var}_M(\mathbb{E}[Y | M]). \quad (3)$$

The second term is the component induced by variation in persistent memory state. If reward corruption changes which memory is written and the paired memories produce different conditional success probabilities, reward uncertainty contributes directly to this term. Section 5 estimates the paired conditional effect and converts it into an explicit corruption-rate variance curve.

This channel is immediate for external-memory agents. No parameter update is required: a mislabeled episode can change the context available to the very next retrieval.

3 F0: DOES THE REWARD BIT CHANGE WRITTEN MEMORY?

We begin with a paired intervention over released WebArena Shopping trajectories from the Salesforce fragility study. A deterministic sampler requests six source trajectories, balanced across original benchmark outcomes. For each trajectory, the task text and full action history are copied byte-for-byte into two ReasoningBank writer calls. One call uses the released success-reflection instruction and the paired call uses the released failure-reflection instruction. DeepSeek-V4-Flash writes memory at temperature zero, and raw provider responses are content-addressed before analysis.

Four of the six requested trajectory pairs return complete memory outputs in both conditions. The two incomplete requests terminate in the failure-label writer before assistant text is produced and are excluded from the paired effect calculation. Among the four complete pairs, every pair changes normalized memory content and the extracted memory-item title set. Token-set Jaccard distance is 0.691 for task 21 and 0.708 for task 22. It is 0.770 for both task 23 and task 25, giving mean distance 0.735. The 4/4 change rate is therefore an existence result conditional on paired completion. Panel (a) of Figure 2 shows the complete pair-level result.

The content changes are strategy-level as well as lexical. For task 21, success-conditioned memory centers on entering the review section and preserving direct quotations. Failure-conditioned memory emphasizes exhaustive pagination and extraction verification. For task 23, the success writer emphasizes evidence collection across pages while the failure writer emphasizes pagination state and broader synonym coverage. To make this structural distinction explicit without introducing another learned judge, we run a deterministic strategy-slot audit over the frozen memory-item titles. A fixed post-ready taxonomy maps titles to navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. The strategy-slot set changes in all four complete pairs; mean slot-set Jaccard distance is 0.493 (range 0.400–0.571). This diagnostic does not claim embedding-level semantic separation, but it shows that the reward-mode effect is not visible only through token overlap. The source trajectory is identical within each pair.

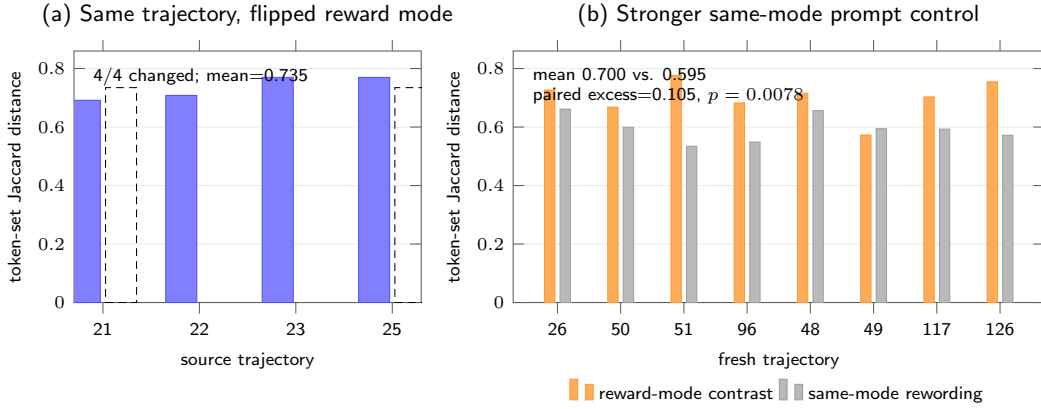


Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

The F0 result establishes the first link of the mechanism: changing the reward-conditioned writer branch changes persistent memory under a fixed experience. Because the branch itself changes the writer instruction, the next section tests the strongest simple alternative explanation: generic sensitivity to prompt wording.

4 RULING OUT GENERIC PROMPT SENSITIVITY

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8} \sum_{i=1}^8 (B_i - W_i). \quad (4)$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

5 C4: DOES THE MEMORY PERTURBATION REACH FUTURE BEHAVIOR?

The write-channel result matters only if the persistent state difference survives reuse. We test this in two stages. The first asks whether paired memories can change an immediate decision under

an identical browser observation. The second measures terminal success under a fully crossed source-memory by future-task design.

5.1 FIXED-STATE ACTION REACH

Four complete F0 pairs are mapped to four held-out Shopping states. Within each state, the future task prompt and browser observation are identical across memory conditions. The policy model and output schema are fixed. The original action-reach experiment contains 12 aligned success-memory versus failure-memory rollouts; 7 pairs choose different structured next actions, giving divergence rate 0.583 and passing the frozen existence falsifier. This establishes an action-level witness: paired memories can change the next decision under the same observation.

We separately stress-test the stronger claim that the first-action distributions are systematically separated across the frozen state set. This test uses eight rollouts per condition on every frozen state. Mean empirical total-variation distance between first-action distributions is 0.344. The 100,000-draw stratified permutation audit gives $p = 0.311$, so this stronger distribution-level test is a negative result: the available state set does not resolve systematic first-action distribution separation. The action-level evidence therefore rests on the paired existence witness (7/12 divergent next actions), not on a population-level distribution claim. Panel (a) of Figure 3 places this negative stress test beside the positive terminal evidence.

5.2 FULLY CROSSED TERMINAL CONFIRMATION

Terminal evaluation uses all four complete F0 source-memory pairs and four future tasks frozen before the first terminal-policy call. Every source pair is crossed with every future task, producing 16 cells. Within a cell, released browser evidence is byte-identical across memory conditions. Doubao Seed 2.0 Mini generates the terminal answer at temperature 0.2, and the original deterministic WebArena string-match evaluator scores success.

The initial frozen run uses three rollouts per memory condition. All 96 primary calls complete. Mean absolute success-rate difference across cells is 0.146 with permutation $p = 0.160$. We then freeze a uniform confirmatory replication over the same complete 4-by-4 matrix before further calls. The confirmatory run uses eight fresh rollouts per condition in every cell. All 256 calls complete with zero provider failures. Mean absolute success-rate difference is 0.15625, exceeding the preregistered 0.15 effect floor, and the 100,000-draw within-cell permutation test gives $p = 0.00074$. Each cell rate moves in increments of 0.125 because each condition has eight rollouts. The prespecified inference target is therefore the aggregate over all 16 frozen cells formed by the four complete F0 source-memory pairs. The individual cells are heterogeneous. Eight have zero observed contrast. Positive failure-minus-success effects occur in six cells, while two cells have negative effects. The two largest cells, source/future pairs 22/388 and 25/387, have signed effects 0.750 and 0.625 and account for 78.2% of the squared-effect mass. We therefore interpret the result as finite distributional sensitivity and a variance channel, not as a stable per-task directional effect. Because the two provider-incomplete source pairs are absent from this matrix, the observed concentration is also conditional on the four completed source pairs; the missing arms prevent estimating how the concentration would change over all six requested sources. Panel (b) of Figure 3 shows the complete signed matrix. F2R1 contains no no-memory terminal arm, so its estimand is the contrast between the two reward-conditioned memories under matched evidence. It establishes that the two written states differ in downstream consequences; it does not establish whether either memory condition is better than omitting memory.

The signed failure-memory minus success-memory effect averages 0.09375, while individual cells have both signs. Variance amplification therefore does not require every reward error to be uniformly harmful. It requires reward-conditioned memory state to alter conditional outcome distributions across runs.

5.3 PLUG-IN VARIANCE DECOMPOSITION UNDER REWARD CORRUPTION

Let $p_S(c)$ and $p_F(c)$ denote the confirmed success probabilities for success-conditioned and failure-conditioned memory in frozen cell c . If reward-label corruption selects the failure-conditioned

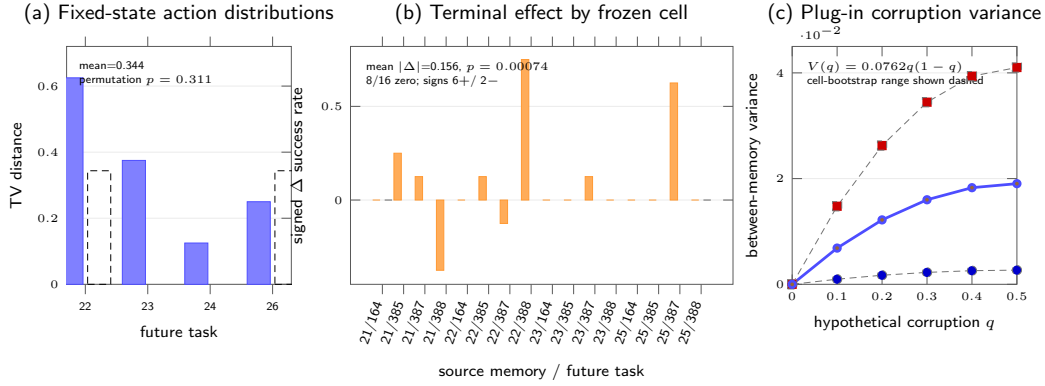


Figure 3: Downstream evidence for C4. (a) Across four frozen states with eight rollouts per memory condition, the larger first-action distributional stress test does not pass its permutation gate ($p = 0.311$). (b) The 16-cell matrix crosses four complete F0 source-memory pairs with four frozen future tasks; signed terminal effects reveal substantial heterogeneity, with aggregate mean absolute difference 0.15625 and $p = 0.00074$. (c) A plug-in decomposition gives the between-memory variance curve for hypothetical reward-label corruption probability q ; dashed curves show the 95% cell-resampling sensitivity range.

memory with probability q , the between-memory success variance in that cell is

$$V_c(q) = q(1-q) [p_F(c) - p_S(c)]^2. \quad (5)$$

This curve is a deterministic plug-in decomposition of the measured conditional effects, not an additional corruption-sweep rollout experiment. Across all 16 confirmatory cells, the mean squared conditional effect is 0.0762. Averaging over cells gives

$$\bar{V}(q) = 0.0762q(1-q). \quad (6)$$

A 100,000-repetition nonparametric bootstrap that resamples the 16 frozen cells gives a descriptive 95% sensitivity interval $[0.0107, 0.1641]$ for the mean squared effect. At $q = 0.5$, the plug-in estimate is 0.0190 with corresponding cell-resampling interval $[0.0027, 0.0410]$. The preregistered within-cell permutation test remains the primary confirmatory significance test. Figure 3(c) shows the plug-in curve together with this finite-cell sensitivity range.

6 RELATED WORK

Reward and judge reliability. JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for learning (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench shows additional brittleness on tool-integrated trajectories (Wang et al., 2026). Our question begins one step later: what persistent state is created when such a terminal label selects a memory-writing objective?

Outcome-conditioned memory writing. ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026), Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023), and Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). These systems establish the writer/reuse paradigm. We do not claim conditional writing is new; we intervene on the label while keeping the source trajectory identical, making writer mode the treatment.

Memory valuation, reuse, and self-improvement variance. Memory Reward Inflation studies how imperfect proxy reward can be amplified as memories are scored and reused (Asadolahi et al., 2026). RoMeRL addresses misleading trajectory-level utility updates through reduced-order memory utility states (Yang et al., 2026). Ye et al. show run-to-run variance, curriculum sensitivity, and memory underspecification (Ye et al., 2026). These closest works cover downstream valuation and

reuse; our residual object is upstream write-time state construction followed by matched downstream action and terminal tests.

Claim boundary. The paper therefore does not claim generic evaluator robustness, universal harm from failure memories, or a new memory architecture. It identifies one causal channel: a reward-label error can select a different memory-construction objective for an otherwise identical episode, and the resulting persistent state can change later behavior under the frozen configuration.

7 LIMITATIONS

The F0 intervention requests six paired memories. Four pairs return complete outputs in both reward conditions. Both incomplete responses occur in the failure-label arm and terminate before assistant text is produced. Receipt-level forensics classify both as the same provider response-state error (`ArkResponseStateError`); neither failed call has a raw memory SHA, while the private provider response identifiers are archived and GET-only recovery returns no assistant text. By contrast, the four completed failure-label calls produce 777–1,296 output tokens. This evidence distinguishes the missing pairs from downstream memory-parser rejection, but it does not rule out a condition-dependent provider completion process. No memory text exists for those arms, so the direction of any resulting selection cannot be estimated. We therefore report the 4/4 write-channel result only among complete paired interventions and do not extrapolate it to all six requested trajectories.

The downstream action evidence has two strengths. The paired existence test records 7 divergent next actions among 12 aligned rollouts. The larger first-action distributional stress test yields mean TV 0.344 with permutation $p = 0.311$, so that stronger population-level action-distribution statement is not statistically resolved by the available state set. Terminal fixed-evidence evidence is stronger: the independently frozen 256-call replication passes its effect and permutation gates. These terminal experiments hold released browser evidence fixed. Source-faithful live browser navigation remains unexecuted because the released Shopping endpoint and reset service are unreachable from both execution hosts. This environment unavailability prevents the live extension and provides no counterevidence about the fixed-evidence effect.

The corruption-rate result is a variance decomposition of the confirmed paired conditional effects. It identifies the between-memory variance induced when reward corruption randomizes which of the two measured memory states is selected. A full end-to-end self-improvement experiment that repeatedly rebuilds complete memory banks under different corruption masks could additionally measure interactions among many corrupted memories.

The memory writer is DeepSeek-V4-Flash and the released source trajectories come from one WebArena Shopping run. Replication across writer families and task domains is required for model-independent or domain-general effect-size claims. Token-set Jaccard distance is a coarse content metric. The post-ready deterministic strategy-slot audit provides a second structural view: all four complete title sets change strategy slots, with mean slot-set Jaccard distance 0.493. This is not an embedding-based semantic analysis, so richer semantic representations remain useful future diagnostics. The downstream action and terminal experiments separately establish that the paired memories are behaviorally consequential. The terminal confirmation also has no no-memory arm; its estimand is failure-conditioned versus success-conditioned memory, not memory versus omission.

8 CONCLUSION

Reward-conditioned memory writing creates a direct path from evaluator error to persistent agent state. Under an identical released trajectory, flipping the reward-conditioned reflection mode changes every complete paired memory, with mean token-set Jaccard distance 0.735. A disjoint eight-trajectory control shows that this separation exceeds stronger semantic-preserving same-mode prompt rewording by 0.105 on average, with exact $p = 0.0078$.

The persistent state difference reaches later behavior. A fixed-state probe changes the next structured action in 7 of 12 aligned pairs. The independently frozen fixed-evidence terminal replication then completes 256 fresh rollouts over the full 4-by-4 source-memory/future-task matrix and finds mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. A plug-in decomposition

of the confirmed conditional effects gives between-memory success variance $0.0762q(1 - q)$ under reward-label corruption probability q .

In the released ReasoningBank/WebArena Shopping configuration studied here, the terminal evaluator is part of the state-update mechanism: its label selects persistent memory and the resulting memory changes later outcome distributions. Within this configuration, reward reliability is therefore a state-reliability requirement whenever outcomes determine what experience is written into persistent memory.

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A F0 REPRODUCTION DETAILS

A.1 RELEASED TRAJECTORY SAMPLE

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 MEMORY CONSTRUCTION INTERVENTION

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

Table 1: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

A.3 PAIR-LEVEL F0 RESULTS

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 2 reports the paired distances used in the frozen sign-flip test.

Table 2: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source-future cells. Each cell receives eight

fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

Table 3: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$ \Delta $
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
Mean absolute difference				0.15625

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the paper treats the result as a variance channel instead of a uniform directional harm effect.

If reward-label corruption selects the failure-conditioned memory with probability q , the induced between-memory variance equals $q(1 - q)\Delta_c^2$ in cell c . The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1 - q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.